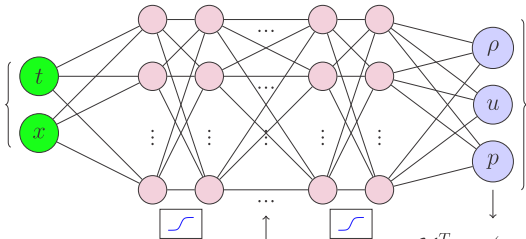


Physical
domain
as input



Automatic differentiation
Physical
variables $\longrightarrow$ $\partial_t, \partial_x, \dots$
as output

$$\mathcal{F}(t, x; \mathbf{u}_\Theta; \partial_x \mathbf{u}_\Theta, \partial_t \mathbf{u}_\Theta, \dots) = 0$$

PDEs

$$\mathcal{L}_\mathcal{R} \rightarrow \hat{\mathcal{L}}_\mathcal{R} := \Lambda(\mathbf{u}_\Theta, \partial_x \mathbf{u}_\Theta) \odot \mathcal{L}_\mathcal{R}$$

$$\mathbf{u}_\Theta^T := (\rho, u, p)$$

$$\mathcal{L}_{IC}$$

$$\mathcal{L} = \omega_{IC} \mathcal{L}_{IC} + \omega_{\mathcal{R}} \hat{\mathcal{L}}_\mathcal{R}$$

No
Yes
End $\leftarrow$ $\mathcal{L} < \epsilon?$